

Chapter 2: Spatial Linear Regression Models

Spatial Econometrics

2.1. Introduction

This chapter introduces popular spatial linear regression models and their estimation through quasi maximum likelihood (QML) method, and GM or GMM method, which include the linear regression models with

- spatial lag dependence (SLD);
- spatial error dependence (SED);
- both SLE and SED;
- spatial Durbin effect, treated as either a direct extension of the SLD model or a direct extension of the SED model.

Tests for covariate effects and for spatial effects are also introduced. Asymptotic properties of the inference procedures introduced are briefly discussed. Some Monte Carlo results are presented to show their finite sample performance. Two examples are presented to illustrate the applications of various model and QML estimation method.

2.2. SED Model

The Model. The linear regression model with spatial error dependence (SED), or the SED model in short, takes the following form:

$$Y_n = X_n\beta + u_n, \quad u_n = \rho W_n u_n + \epsilon_n, \quad (1)$$

where the SED is specified as a **spatial autoregressive** (SAR) process. It can also be of the **spatial moving average** (SMA) form: $u_n = \rho W_n \epsilon_n + \epsilon_n$,

- Y_n : $n \times 1$ vector of observations on n spatial units,
- X_n : an $n \times k$ matrix containing the values of k regressors,
- W_n : $n \times n$ matrix summarizing interactions among n spatial units, called the **spatial weight matrix** or the **spatial interaction matrix**,
- ϵ_n : $n \times 1$ vector of independent and identically distributed (iid) idiosyncratic errors with mean zero and variance σ^2 ,
- ρ : the **spatial error parameter**,
- β : $k \times 1$ vector of regression coefficients.

Durbin-SED Model. The model (1) can be extended by adding a spatial Durbin term $W_n X_n^*$, where X_n^* contains **some** regressors, referred to in this course as **Durbin-SED model**:

$$Y_n = X_n \beta + W_n X_n^* \delta + u_n, \quad u_n = \rho W_n u_n + \epsilon_n. \quad (2)$$

The term '**spatial Durbin model**' was first appeared in Anselin (1988) for its analogy with Durbin (1960) for time series. See also Elhorst (2014, p.7).

- Elhorst (2014) interprets the spatial Durbin effect as the *exogenous interaction effects*, where the dependent variable of a particular unit depends on independent variables of other units.
- By defining a new regressor matrix $\mathbb{X}_n = [X_n, W_n X_n^*]$, and a new vector of regression coefficients $\beta = (\beta', \delta')'$, the Durbin-SED model has the same form as SED model, and hence the inference methods remain the same.
- However, a problem of particular interest is to test the existence of spatial Durbin effect, i.e., testing $H_0: \delta = 0$, which can be carried out based on the usual tests for covariate effects, to be presented latter.

2.2.1. QML estimation.

For QML estimation of spatial models, the following results are useful:

(i) An n -dimensional random vector $\mathbf{Y}$ is said to have a multivariate normal distribution with mean $\boldsymbol{\mu}$ and variance-covariance (VC) matrix $\boldsymbol{\Sigma}$, denoted as $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, if its joint pdf takes the form:

$$\frac{1}{(2\pi)^{n/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \right). \quad (3)$$

where $|\cdot|$ denotes the determinant of a square matrix.

(ii) For two $n \times n$ matrices A and B , and a scalar c :

$$(a) |A'| = |A|, \quad (b) |A^{-1}| = |A|^{-1}, \quad (c) |cA| = c^n |A|, \quad (d) |AB| = |A||B|;$$

(iii) For a matrix function $A(\rho)$ of scalar ρ , positive definite (p.d.),

$$(a) \frac{\partial}{\partial \rho} A(\rho)^{-1} = -A(\rho)^{-1} \left[\frac{\partial}{\partial \rho} A(\rho) \right] A(\rho)^{-1}, \quad \text{letting } \text{tr}(\cdot) = \text{trace of a matrix},$$

$$(b) \frac{\partial}{\partial \rho} \log |A(\rho)| = \text{tr} \left[A(\rho)^{-1} \frac{\partial}{\partial \rho} A(\rho) \right], \quad \text{see Horn and Johnson (1985).}$$

Now, define $B_n(\rho) = I_n - \rho W_n$. Then, $u_n = B_n^{-1}(\rho)\epsilon_n$, and

$$\Sigma_n = \text{Var}(u_n) = \sigma^2 B_n^{-1}(\rho) B_n'^{-1}(\rho),$$

The **quasi** Gaussian loglikelihood function of $\theta = (\beta', \sigma^2, \rho)'$ for SED or Durbin-SED model, **as if** $\epsilon_{n,i}$ are iid normal, is given by,

$$\ell_n(\theta) = -\frac{n}{2} \log(2\pi\sigma^2) + \log |B_n(\rho)| - \frac{1}{2\sigma^2} (Y_n - X_n\beta)' B_n'(\rho) B_n(\rho) (Y_n - X_n\beta). \quad (4)$$

Maximizing $\ell_n(\theta)$ gives the MLE $\hat{\theta}_n$ of θ if the errors are indeed Gaussian, otherwise the QMLE. Maximizing $\ell_n(\theta)$ is equivalent to solving $S_n(\theta) = 0$, where the score function has the form:

$$S_n(\theta) = \frac{\partial}{\partial \theta} \ell_n(\theta) = \begin{cases} \frac{1}{\sigma^2} X_n' B_n'(\rho) B_n(\rho) u_n(\beta), \\ \frac{1}{2\sigma^4} u_n'(\beta) B_n'(\rho) B_n(\rho) u_n(\beta) - \frac{n}{2\sigma^2}, \\ \frac{1}{\sigma^2} u_n'(\beta) B_n'(\rho) W_n u_n(\beta) - \text{tr}[G_n(\rho)], \end{cases} \quad (5)$$

where $u_n(\beta) = Y_n - X_n\beta$, and $G_n(\rho) = W_n B_n^{-1}(\rho)$.

The QML estimation process can be simplified:

- Solving the first two sets of equations of (5) for a given ρ gives the constrained QMLEs of β and σ^2 ,

$$\tilde{\beta}_n(\rho) = [X_n' B_n'(\rho) B_n(\rho) X_n]^{-1} X_n' B_n'(\rho) B_n(\rho) Y_n, \text{ and} \quad (6)$$

$$\tilde{\sigma}_n^2(\rho) = \frac{1}{n} Y_n' B_n'(\rho) M_n(\rho) B_n(\rho) Y_n, \quad (7)$$

where $M_n(\rho) = I_n - B_n(\rho) X_n [X_n' B_n'(\rho) B_n(\rho) X_n]^{-1} X_n' B_n'(\rho)$.

- The concentrated log-likelihood function for ρ upon substituting the constrained QMLEs $\tilde{\beta}_n(\rho)$ and $\tilde{\sigma}_n^2(\rho)$ into $\ell(\theta)$:

$$\ell_n^c(\rho) = -\frac{n}{2} [\log(2\pi) + 1] + \log |B_n(\rho)| - \frac{n}{2} \log(\tilde{\sigma}_n^2(\rho)). \quad (8)$$

- Maximising $\ell_n^c(\rho)$ gives the unconstrained QMLE $\hat{\rho}_n$ of ρ ,
- which upon substitutions gives the unconstrained QMLEs of β and σ^2 as, $\hat{\beta}_n \equiv \tilde{\beta}_n(\hat{\rho}_n)$ and $\hat{\sigma}_n^2 \equiv \tilde{\sigma}_n^2(\hat{\rho}_n)$.
- Thus, the QMLE of the full parameter vector θ is $\hat{\theta}_n = (\hat{\beta}_n', \hat{\sigma}_n^2, \hat{\rho}_n)'$.

Asymptotic properties of QMLE. Let θ_0 be the true value of the parameter vector θ that generates the data; $E(\cdot)$ and $\text{Var}(\cdot)$ correspond to θ_0 ; and $B_n \equiv B_n(\lambda_0)$, $G_n \equiv G_n(\lambda_0)$, etc. Liu and Yang (2015a) show that under regularity conditions: $\hat{\theta}_n \xrightarrow{P} \theta_0$, and

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{D} N(0, \lim_{n \rightarrow \infty} n \mathcal{J}_n^{-1} \mathcal{I}_n \mathcal{J}_n^{-1}), \quad (9)$$

where $\mathcal{J}_n = -E[\frac{\partial}{\partial \theta} S_n(\theta_0)]$ and $\mathcal{I}_n = \text{Var}[S_n(\theta_0)]$,

$$\mathcal{J}_n = \begin{pmatrix} \frac{1}{\sigma_0^2} X_n' B_n' B_n X_n & 0 & 0 \\ \sim & \frac{n}{2\sigma_0^4} & \frac{1}{\sigma_0^2} \text{tr}(G_n) \\ \sim & \sim & \text{tr}(G_n^s G_n) \end{pmatrix}, \quad (10)$$

$$\mathcal{I}_n = \begin{pmatrix} \frac{1}{\sigma_0^2} X_n' B_n' B_n X_n & \frac{1}{2\sigma_0^3} \gamma_0 X_n' B_n' \iota_n & \frac{1}{\sigma_0} \gamma_0 X_n' B_n' g_n \\ \sim & \frac{n}{4\sigma_0^4} (\kappa_0 + 2) & \frac{1}{2\sigma_0^2} (\kappa_0 + 2) \text{tr}(G_n) \\ \sim & \sim & \kappa_0 g_n' g_n + \text{tr}(G_n^s G_n) \end{pmatrix}, \quad (11)$$

ι_n is a vector of ones, γ_0 and κ_0 are the measures of skewness and excess kurtosis of $\epsilon_{n,i}$, $g_n = \text{diagv}(G_n)$, $G_n = G_n(\rho_0)$, and $G_n^s = G_n + G_n'$.

Remarks:

- $\text{diagv}(G_n)$ forms a column vector by the diagonal elements of G_n .
- Note that in deriving the last component (the ρ - ρ component) of $\mathcal{J}_n$, we have used the following matrix identity (given above):

$$\frac{\partial}{\partial \rho} B_n^{-1}(\rho) = -B_n^{-1}(\rho) \left[\frac{\partial}{\partial \rho} B_n(\rho) \right] B_n^{-1}(\rho). \quad (12)$$

- Clearly, when $\epsilon_{n,i}$ are iid normal, $\gamma_0 = \kappa_0 = 0$, and the asymptotic result reduces to $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{D} N(0, \lim_{n \rightarrow \infty} n\mathcal{J}_n^{-1})$.

For deriving the expression for $\mathcal{I}_n = \text{Var}[S_n(\theta_0)]$, we have by (5),

$$S_n(\theta_0) = \begin{cases} \frac{1}{\sigma_0^2} X_n' B_n' \epsilon_n, \\ \frac{1}{2\sigma_0^4} \epsilon_n' \epsilon_n - \frac{n}{2\sigma_0^2}, \\ \frac{1}{\sigma_0^2} \epsilon_n' G_n \epsilon_n - \text{tr}(G_n), \end{cases}$$

of which the elements are linear or quadratic forms in ϵ_n .

Using the following easily proved results, $\text{Var}[S_n(\theta_0)]$ is readily obtained.

Let A_n and B_n be $n \times n$ non-stochastic matrices and c_n be $n \times 1$ non-stochastic vector. For $n \times 1$ random vector ϵ_n of iid elements with mean 0, variance σ_0^2 , skewness γ_0 , and finite excess kurtosis κ_0 , we have

$$(i) \text{Cov}(c_n' \epsilon_n, \epsilon_n' A_n \epsilon_n) = \sigma_0^3 \gamma_0 c_n' a_n,$$

$$(ii) \text{Cov}(\epsilon_n' A_n \epsilon_n, \epsilon_n' B_n \epsilon_n) = \sigma_0^4 \kappa_0 a_n' b_n + \sigma_0^4 \text{tr}(A_n B_n^s),$$

where $a_n = \text{diagv}(A_n)$, $b_n = \text{diagv}(B_n)$, and $B_n^s = B_n' + B_n$.

Asymptotic inferences. Based on the above results, estimating

- 1 $\mathcal{J}_n$ by $\hat{\mathcal{J}}_n = \mathcal{J}_n(\hat{\theta}_n)$,
- 2 $\mathcal{I}_n$ by $\hat{\mathcal{I}}_n = \mathcal{I}(\hat{\theta}_n)$, and
- 3 γ_0 and κ_0 in $\mathcal{I}_n$ by the sample skewness and excess kurtosis of the estimated errors: $\hat{\epsilon}_n = \epsilon_n(\hat{\rho}_n, \hat{\beta}_n) = B_n(\hat{\rho}_n)(Y_n - X_n \hat{\beta}_n)$,

we obtained robust variance-covariance (VC) matrix, and thus robust standard errors of $\hat{\theta}_n$ and robust t -ratios for θ_0 .

2.2.2. GLS-GM estimation

A hybrid method, combination of generalized least squares (GLS) and generalized moments (GM), for estimating the SED model is introduced by Kelejian and Prucha (1999). The ideas are

- first to develop GM estimators $\tilde{\rho}_n$ and $\tilde{\sigma}_n^2$ for ρ and σ^2 , based on a consistent 'predictor' $\tilde{u}_n$ for u_n ;
- then to estimate $\Sigma_n = \text{Var}(u_n)$ by $\tilde{\Sigma}_n = \tilde{\sigma}_n^2 B_n^{-1}(\tilde{\rho}_n) B_n'^{-1}(\tilde{\rho}_n)$, leading to a feasible GLS estimate for β as

$$\tilde{\beta}_n = (X_n' \tilde{\Sigma}_n^{-1} X_n)^{-1} X_n' \tilde{\Sigma}_n^{-1} Y_n;$$

- Under typical conditions, the GLS estimator of β based on Σ_n is consistent and asymptotically normal;
- Under additional conditions, e.g., $\tilde{\rho}_n$ and $\tilde{\sigma}_n^2$ are consistent, the feasible GLS estimator is asymptotically equivalent to the GLS estimator, and thus is also consistent and asymptotically normal.

GM estimation. The generalized moments (GM) estimation of ρ and σ^2 is based on the following three moment conditions:

$$\begin{aligned} E\left(\frac{1}{n}\epsilon'_n\epsilon_n\right) &= \sigma^2, \\ E\left(\frac{1}{n}\epsilon'_n W'_n W_n \epsilon_n\right) &= \sigma^2 n^{-1} \text{tr}(W'_n W_n), \\ E\left(\frac{1}{n}\epsilon'_n W'_n \epsilon_n\right) &= 0. \end{aligned}$$

By $\epsilon_n = B_n(\rho)u_n$, we obtain the following sample moment conditions:

$$\mathbf{g}_n(\rho, \sigma^2) = \begin{cases} \frac{1}{n}\tilde{u}'_n B'_n(\rho) B_n(\rho) \tilde{u}_n - \sigma^2, \\ \frac{1}{n}\tilde{u}'_n B'_n(\rho) W'_n W_n B_n(\rho) \tilde{u}_n - \frac{1}{n}\sigma^2 \text{tr}(W'_n W_n), \\ \frac{1}{n}\tilde{u}'_n B'_n(\rho) W'_n B_n(\rho) \tilde{u}_n. \end{cases}$$

The GM estimators of ρ and σ^2 are thus:

$$(\tilde{\rho}_n, \tilde{\sigma}_n^2) = \text{argmin} [\mathbf{g}'_n(\rho, \sigma^2) \mathbf{g}_n(\rho, \sigma^2)],$$

Remark: $\tilde{u}_n$ can be the ordinary least squares (OLS) residuals, i.e., the residuals obtained from regressing Y_n on X_n (Kelejian and Prucha, 1999).

2.2.3. Testing statistical hypothesis

Statistical inferences can be carried out based on the three classical inference procedures, namely, Wald, LR, and LM (Score).

Tests for covariates effects. From the asymptotic results (9)-(11) for the QMLEs, it is easy to see that $\hat{\beta}_n$ is asymptotically $N(\beta_0, \sigma_0^2(X_n' B_n' B_n X_n)^{-1})$. Thus, the inference for $c_0' \beta_0$ is carried out based on the following t -ratio:

$$t_{\text{SED}}(\beta_0) = \frac{c_0' \hat{\beta}_n - c_0' \beta_0}{\sqrt{\hat{\sigma}_n^2 c_0' (X_n' \hat{B}_n' \hat{B}_n X_n)^{-1} c_0}}, \quad (13)$$

where c_0 represents a linear contrast and $\hat{B}_n = I_n - \hat{\rho}_n W_n$.

- The t -ratio, t_{SED} , is asymptotically $N(0, 1)$, and hence inferences for β_0 are carried out by referring to the $N(0, 1)$ critical values.
- By construction, this test is robust against nonnormality of the errors.
- However, its finite sample performance may be poor. Liu and Yang (2015b) presented improved tests based on bias-correction.

A multivariate version of (13) for testing the whole vector or a subset of β_0 :

$$F_{\text{SED}}(\beta_0) = (\hat{\beta}_n - \beta_0)' C_0' [\hat{\sigma}_n^2 C_0' (X_n' \hat{B}_n' \hat{B}_n X_n)^{-1} C_0]^{-1} C_0 (\hat{\beta}_n - \beta_0), \quad (14)$$

where C_0 is $p \times k$ represents $p(\leq k)$ linear contrasts. The asymptotic null distribution of the statistic $F_{\text{SED}}(\beta_0)$ is χ_p^2 .

Tests for spatial effect. Of particular interest is the test of $H_0: \rho = 0$. Based on (9)-(11), a t -ratio for inference for ρ is constructed:

$$t_{\text{SED}}(\rho_0) = \frac{\hat{\rho}_n - \rho_0}{\sqrt{(\hat{\mathcal{J}}_n^{-1} \hat{\mathcal{I}}_n \hat{\mathcal{J}}_n^{-1})_{\rho\rho}}}, \quad (15)$$

where $\hat{\mathcal{J}}_n$ and $\hat{\mathcal{I}}_n$ are the plug-in estimates of $\mathcal{J}_n$ and $\mathcal{I}_n$, and $(\cdot)_{\rho\rho}$ denotes the ρ - ρ element of the corresponding matrix.

- Again, this test is robust against nonnormality.
- However, its finite sample property is not clear.
- An LR test can be constructed as well, but its robustness is unclear.

LM tests for Spatial Error Dependence.

Among the three classical tests, the LM is of particular interest as its implementation requires only the estimation of the null model.

- Its advantage is clear if the null hypothesis specifies that $\rho = 0$, and hence the implementation of the LM test requires only OLS estimates.
- Many tests of this type are available in the literature.

Moran's I test: from Moran (1950), extended by Cliff and Ord (1972):

$$I = \frac{\tilde{\epsilon}_n' W_n \tilde{\epsilon}_n}{\tilde{\epsilon}_n' \tilde{\epsilon}_n}, \quad (16)$$

where $\tilde{\epsilon}_n$ is a vector of OLS residuals, from regressing Y_n on X_n . If ϵ_n is normal, then under $H_0: \rho = 0$, I is asymptotically normally distributed with mean and variance:

$$\begin{aligned} E(I) &= \frac{1}{n-k} \text{tr}(M_n W_n), \\ \text{Var}(I) &= \frac{\text{tr}(M_n W_n M_n W_n') + \text{tr}((M_n W_n)^2) - \frac{2}{n-k} [\text{tr}(M_n W_n)]^2}{(n-k)(n-k+2)}. \end{aligned}$$

Here $M_n = I_n - X_n(X_n' X_n)^{-1} X_n'$ and I_n is an $n \times n$ identity matrix.

EI-based LM test: by Burrridge (1980) for testing $H_0 : \rho = 0$, based on the expected information (EI) matrix $\mathcal{I}_n$ given in (11):

$$\text{LME}_{\text{SED}} = \frac{n}{\sqrt{S_0}} \frac{\tilde{\epsilon}'_n W_n \tilde{\epsilon}_n}{\tilde{\epsilon}'_n \tilde{\epsilon}_n}, \quad (17)$$

where $S_0 = \text{tr}(W'_n W_n + W_n^2)$, having asymptotic null distribution $N(0, 1)$.

OPG-based LM test: Born and Breitung (2011) derived an outer-product-of-gradients (OPG) variant of Burrridge's LM test:

$$\text{LMO}_{\text{SED}} = \frac{\tilde{\epsilon}'_n W_n \tilde{\epsilon}_n}{\sqrt{(\tilde{\epsilon}_n \odot \tilde{\epsilon}_n)'(\tilde{\xi}_n \odot \tilde{\xi}_n)}}, \quad (18)$$

where $\odot$ denotes the Hadamard product, $\tilde{\xi}_n = (W_n^l + W_n^{u'})\tilde{\epsilon}_n$, W_n^l and W_n^u are the lower and upper triangular matrices of W_n , and $\text{LMO}_{\text{OPG}}|_{H_0} \xrightarrow{D} N(0, 1)$.

- LME_{SED} is robust against unknown heteroskedasticity.
- Baltagi and Yang (2013a) commented both LME_{SED} and LMO_{SED} may have poor finite sample as they are neither centered, nor standardized.
- They introduced a standardized versions of these tests which have much better finite sample performance (see Chapter 3 of the lecture notes).

2.2.4. Finite sample performance

Liu and Yang (2015a) demonstrate based on Monte Carlo experiments:

- $\hat{\rho}_n$ can be seriously downward biased but the bias of $\hat{\rho}_n$ does not spillover much to $\hat{\beta}_n$, in terms of consistency and finite sample bias;
- However, it does spill over to the estimate of $\text{Var}(\hat{\beta}_n)$:
 - first, the downward bias of $\hat{\rho}_n$ causes $\hat{\sigma}_n^2$ to be downward biased,
 - second, severe bias of $\hat{\rho}_n$ may cause $X_n' \hat{B}_n' \hat{B}_n X_n$ to be severely biased for the estimation of $X_n' B_n' B_n X_n$ as seen from the expression:

$$\begin{aligned} X_n' \hat{B}_n' \hat{B}_n X_n &= X_n' B_n' B_n X_n - (\hat{\rho}_n - \rho_0) X_n' (W_n' B_n + B_n' W_n) X_n \\ &\quad + (\hat{\rho}_n - \rho_0)^2 X_n' W_n' W_n X_n, \end{aligned}$$

- e.g., if $X_n' (W_n' B_n + B_n' W_n) X_n \geq 0$, then $X_n' \hat{B}_n' \hat{B}_n X_n$ overestimates $X_n' B_n' B_n X_n$, and hence, $\hat{\sigma}_n^2 c_0' (X_n' \hat{B}_n' \hat{B}_n X_n)^{-1} c_0$ underestimates $\text{Var}(c_0' \hat{\beta}_n)$.
- This makes $t_{\text{SED}}(\beta_0)$ much more variable than $N(0, 1)$ and inferences for β_0 based on t_{SED} defined in (13) unreliable.
- Monte Carlo results in Liu and Yang (2015b) confirm this point.

2.2.5 Empirical illustrations

Neighborhood Crime. In illustrating the applications of spatial cross-sectional models, Anselin (1988, p.187) used the neighborhood crime data corresponding to 49 contiguous neighborhood in Columbus, Ohio, in 1980. These neighborhood correspond to census tracts, where

- **Crime:** the response variable pertaining to the combined total of residential burglaries and vehicle thefts per thousand household in the neighborhood.
- **Income** and housing values (**House**), are the explanatory variables are both in thousand dollars.
- A dummy variable **East** indicates whether the 'neighborhood' in the east or west of a main north-south transportation axis.

Table 2.1. Estimation of SED Model: Neighborhood Crime

	QMLE	se	t-Ratio	rse	rt-Ratio
constant	59.8924	5.3662	11.1611	5.3662	11.1611
income	-0.9413	0.3306	-2.8477	0.3306	-2.8477
hvalue	-0.3023	0.0905	-3.3407	0.0905	-3.3407
σ^2	95.5737	19.8735	4.8091	27.1596	3.5190
ρ	0.5618	0.1339	4.1963	0.1343	4.1835

Boston House Price. The data, from Harrison and Rubinfeld (1978), corrected and augmented with longitude and latitude by Gilley and Pace (1996), contains 506 observations (1 observation per census tract) from Boston Metropolitan Statistical Area. The variables are:

- MEDV: the median value (corrected) of owner-occupied homes in 1000's;
- crime: per capita crime rate by town;
- zoning: proportion of residential land zoned for lots over 25,000 square feet;
- industry: proportion of non-retail business acres per town;
- charlesr: Charles River dummy variable (= 1 if tract bounds river; 0 o.w.);
- nox: nitric oxides concentration (parts per 10 million);
- room: average number of rooms per dwelling;
- houseage: proportion of owner-occupied units built prior to 1940;
- distance: weighted distances to five Boston employment centres;
- access: index of accessibility to radial highways;
- taxrate: full-value property-tax rate per 10,000;
- ptratio: pupil-teacher ratio by town;
- blackpop: $1000(Bk - 0.63)^2$ where Bk is the proportion of blacks by town;
- lowclass: lower status of the population proportion.

The spatial weight matrix is constructed using the Euclidean distance in terms of longitude and latitude. A threshold distance e.g., 0.05, is chosen, which gives a W_n matrix with 19.08% non-zero elements.

Table 2.2. Estimation of SED Model: Boston House Price (MEDV)

	QMLE	se	t-Ratio	rse	rt-Ratio
constant	29.6250	5.4956	5.3907	5.4956	5.3907
crime	-0.1318	0.0276	-4.7693	0.0276	-4.7693
zoning	0.0379	0.0141	2.6887	0.0141	2.6887
industry	-0.0139	0.0729	-0.1909	0.0729	-0.1909
charlesr	-0.4975	0.8794	-0.5658	0.8794	-0.5658
noxsq	-19.2666	5.2686	-3.6568	5.2686	-3.6568
rooms2	4.2812	0.3643	11.7516	0.3643	11.7516
houseage	-0.0259	0.0139	-1.8604	0.0139	-1.8604
distance	-1.6095	0.3021	-5.3276	0.3021	-5.3276
access	0.3174	0.0767	4.1407	0.0767	4.1407
taxrate	-0.0130	0.0036	-3.6137	0.0036	-3.6137
ptratio	-0.6143	0.1523	-4.0344	0.1523	-4.0344
blackpop	0.0106	0.0031	3.4261	0.0031	3.4261
lowclass	-0.4270	0.0514	-8.2999	0.0514	-8.2999
σ^2	14.9219	0.9697	15.3874	1.7700	8.4306
ρ	0.6947	0.0420	16.5461	0.0420	16.5332

2.3. SLD Model

The model. The spatial lag dependence (SLD) model takes the form:

$$Y_n = \lambda W_n Y_n + X_n \beta + \epsilon_n, \quad (19)$$

where Y_n , X_n , and W_n are as in (1). The errors $\epsilon_{n,i}$ are iid(0, σ^2).

- The term $\lambda W_n Y_n$ says that the dependent variable of a specific spatial unit may depend on the dependent variables of other spatial units.
- This model captures the possible **endogenous interaction effects**.
- A major difference between SLD and SED models is that the spatial interactions in SED model changes only the variance of Y_n , whereas in SLD model, it changes both the mean and the variance of Y_n :

$$\text{SED model: } E(Y_n) = X_n \beta, \quad \text{Var}(Y_n) = \sigma^2 B_n^{-1}(\rho) B_n'^{-1}(\rho),$$

$$\text{SLD model: } E(Y_n) = A_n^{-1}(\lambda) X_n \beta, \quad \text{Var}(Y_n) = \sigma^2 A_n^{-1}(\lambda) A_n'^{-1}(\lambda),$$

where $A_n(\lambda) = I_n - \lambda W_n$. **This makes the SLD model more popular.**

Durbin-SLD model. Similar to the SED model, the SLD model (19) can also be extended by adding a spatial Durbin term $W_n X_n^*$, where X_n^* some regressors (other than the constant term), giving the so-called *Durbin-SLD model* (see Elhorst, 2014, p.7). The model has the form

$$Y_n = \lambda W_n Y_n + X_n \beta + W_n X_n^* \delta + \epsilon_n. \quad (20)$$

By defining a new regressor matrix $\mathbb{X} = [X_n, W_n X_n^*]$, and a new vector of regression coefficients $\beta = (\beta', \delta')'$,

- the spatial Durbin model (20) has an identical form as the regular the SLD model, and hence all the estimation and inference methods are kept the same.
- However, a problem of particular interest, as in the spatial Durbin error model, is to test the existence of spatial Durbin effect, i.e., testing $H_0: \delta = 0$, which can be carried out based on the tests for covariate effects.

2.3.1. QML estimation

Using the matrix $A_n(\lambda)$, the model (19) has a reduced form:

$$A_n(\lambda)Y_n = X_n\beta + \epsilon_n.$$

Assuming $\epsilon_{n,i}$ are iid $N(0, \sigma^2)$, the joint pdf of ϵ_n is

$$(2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}\epsilon_n'\epsilon_n\right),$$

which directly translates into the joint pdf of Y_n as the **Jacobian** of the inverse transformation ($\epsilon_n \rightarrow Y_n$) is $|A_n(\lambda)|$. Thus, the log-likelihood function of $\theta = (\beta', \sigma^2, \lambda)'$ is

$$\begin{aligned}\ell_n(\theta) &= -\frac{n}{2} \log(2\pi\sigma^2) + \log |A_n(\lambda)| \\ &\quad - \frac{1}{2\sigma^2} [A_n(\lambda)Y_n - X_n\beta]' [A_n(\lambda)Y_n - X_n\beta].\end{aligned}\tag{21}$$

Maximizing $\ell_n(\theta)$ gives the MLE $\hat{\theta}_n$ of θ if $\epsilon_{n,i}$ are truly iid $N(0, \sigma^2)$, otherwise the QMLE if $\epsilon_{n,i}$ are merely iid(0, σ^2).

Similar to the case of SED model, the MLE or QMLE $\hat{\theta}_n$ can be obtained by solving the score-based estimating equations $S_n(\theta) = 0$, where

$$S_n(\theta) = \frac{\partial}{\partial \theta} \ell_n(\theta) = \begin{cases} \frac{1}{\sigma^2} X_n' \epsilon_n(\beta, \lambda), \\ \frac{1}{2\sigma^4} \epsilon_n'(\beta, \lambda) \epsilon_n(\beta, \lambda) - \frac{n}{2\sigma^2}, \\ \frac{1}{\sigma^2} Y_n' W_n' \epsilon_n(\beta, \lambda) - \text{tr}[G_n(\lambda)], \end{cases} \quad (22)$$

$\epsilon_n(\beta, \lambda) = A_n(\lambda) Y_n - X_n \beta$, and $G_n(\lambda) = W_n A_n^{-1}(\lambda)$. Thus, the process of obtaining $\hat{\theta}_n$ can be simplified:

- Solving the first two sets of equations in (22) for a given λ , we obtain the constrained QMLEs for β and σ^2 , letting $M_n = I_n - X_n(X_n' X_n)^{-1} X_n'$:

$$\tilde{\beta}_n(\lambda) = (X_n' X_n)^{-1} X_n' A_n(\lambda) Y_n, \quad \tilde{\sigma}_n^2(\lambda) = \frac{1}{n} Y_n' A_n(\lambda) M_n A_n(\lambda) Y_n. \quad (23)$$

- Substituting $\tilde{\beta}_n(\lambda)$ and $\tilde{\sigma}_n^2(\lambda)$ back into (21) for β and σ^2 , we obtain the partially maximized or the concentrated loglikelihood of λ :

$$\ell_n^c(\lambda) = -\frac{n}{2} [\log(2\pi) + 1] - \frac{n}{2} \log \hat{\sigma}_n^2(\lambda) + \log |A_n(\lambda)|. \quad (24)$$

- Maximizing $\ell_n^c(\lambda)$ gives $\hat{\lambda}_n$, $\hat{\beta}_n \equiv \tilde{\beta}_n(\hat{\lambda}_n)$, $\hat{\sigma}_n^2 \equiv \tilde{\sigma}_n^2(\hat{\lambda}_n)$, and hence $\hat{\theta}_n$.

Asymptotic properties of QMLE. Using a set of notation similar to that of Section 2.2, e.g., θ_0 , $E(\cdot)$, $\text{Var}(\cdot)$, $A_n \equiv A_n(\lambda_0)$, and $G_n \equiv G_n(\lambda_0)$. Under some regularity conditions, Lee (2004) shows that

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{D} N(\theta_0, \lim_{n \rightarrow \infty} n \mathcal{J}_n^{-1} \mathcal{I}_n \mathcal{J}_n^{-1}), \quad (25)$$

where $\mathcal{J}_n = -E[\frac{\partial}{\partial \theta'} S_n(\theta_0)]$ and $\mathcal{I}_n = \text{Var}[S_n(\theta_0)]$, having expressions:

$$\mathcal{J}_n = \begin{pmatrix} \frac{1}{\sigma_0^2} X_n' X_n & 0 & \frac{1}{\sigma_0} X_n' \eta_n \\ 0 & \frac{n}{2\sigma_0^4} & \frac{1}{\sigma_0^2} \text{tr}(G_n) \\ \frac{1}{\sigma_0} \eta_n' X_n & \frac{1}{\sigma_0^2} \text{tr}(G_n) & \eta_n' \eta_n + \text{tr}(G_n^s G_n) \end{pmatrix} \quad (26)$$

$$\mathcal{I}_n = \begin{pmatrix} \frac{1}{\sigma_0^2} X_n' X_n & \frac{1}{2\sigma_0^3} \gamma_0 X_n' \iota_n & \frac{1}{\sigma_0} X_n' \eta_n + \frac{1}{\sigma_0} \gamma_0 X_n' g_n \\ \sim & \frac{n}{2\sigma_0^4} + \frac{n}{4\sigma_0^4} \kappa_0 & \frac{1}{\sigma_0^2} \text{tr}(G_n) + \frac{1}{2\sigma_0^2} \gamma_0 \iota_n' \eta_n + \frac{1}{2\sigma_0^2} \kappa_0 \text{tr}(G_n) \\ \sim & \sim & \eta_n' \eta_n + \text{tr}(G_n^s G_n) + \kappa_0 g_n' g_n + 2\gamma_0 g_n' \eta_n \end{pmatrix} \quad (27)$$

where $g_n = \text{diagv}(G_n)$, $G_n^s = G_n + G_n'$, $\eta_n = \sigma_0^{-1} G_n X_n \beta_0$, ι_n denotes an $n \times 1$ vector of ones, and γ_0 and κ_0 skewness and kurtosis of $\epsilon_{n,i}$.

The results (25)-(27) provide QML-based statistical inferences concerning the parameters of the SLD model.

- Again, $\mathcal{J}_n$ can be estimated by the plug-in estimator $\mathcal{J}_n(\hat{\theta}_n)$, or simply by its sample counterpart $-\frac{\partial}{\partial \theta} \mathbf{S}_n(\theta)|_{\theta=\hat{\theta}_n}$.
- The variance of the score $\mathcal{I}_n$ can be consistently estimated by the plug-in method, i.e., plugging
 - $\hat{\theta}_n$ in $\mathcal{I}_n$ for θ_0 , and
 - $\hat{\gamma}_n$ and $\hat{\kappa}_n$ for γ_0 and κ_0 ,
 - where $\hat{\gamma}_n$ and $\hat{\kappa}_n$ are the sample skewness and excess kurtosis of the estimated errors $\hat{\epsilon}_n = \epsilon_n(\hat{\beta}_n, \hat{\lambda}_n) = A_n(\hat{\lambda}_n) Y_n - X_n \hat{\beta}_n$.
- These results give the estimates of robust standard errors of the QMLE $\hat{\theta}_n$ – robust against nonnormality of the error distribution.
- When it is known that ϵ_n is normally distributed, then $\mathcal{I}_n = \mathcal{J}_n$, and $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{D} N(\theta_0, \lim_{n \rightarrow \infty} n \mathcal{J}_n^{-1})$.

2.3.2. GMM estimation

The added regressor $W_n Y_n$ in the SLD model (19) is endogenous, as

$$E[(W_n Y_n) \epsilon_n'] = \sigma_0^2 G_n(\lambda_0) \neq 0,$$

i.e., the elements of $W_n Y_n$ are correlated with the elements of ϵ_n . This can easily be seen from the expression $W_n Y_n = G_n(\lambda_0) X_n \beta_0 + G_n(\lambda_0) \epsilon_n$, which follows from the reduced form of (19): $Y_n = A_n^{-1}(\lambda_0) X_n \beta + A_n^{-1}(\lambda_0) \epsilon_n$.

- Therefore, the OLS estimate of $\delta = (\beta', \lambda)'$ is not consistent.
- If W_n is row normalized and $|\lambda| < 1$, then $(I_n - \lambda W_n)^{-1} = \sum_{i=0}^{\infty} \lambda^i W_n^i$,
- $\Rightarrow$ **instrumental variables (IVs) for $W_n Y_n$:** $W_n X_n, W_n^2 X_n, \dots$,
- $\Rightarrow$ **linear moment functions:** $[X_n, Q_n]' \epsilon_n(\delta)$,
where $\epsilon_n(\delta) = A_n(\lambda) Y_n - X_n \beta$, and Q_n contains the columns of $W_n X_n, W_n^2 X_n, \dots$, which are linearly independent of the columns of X_n .
- For an $n \times n$ matrix P_n with $\text{tr}(P_n) = 0$, $E(\epsilon_n' P_n \epsilon_n) = \text{tr}(P_n E(\epsilon_n \epsilon_n')) = 0$,
- $\Rightarrow$ **quadratic moment functions:** $\epsilon_n'(\delta) P_{jn} \epsilon_n(\delta)$, $j = 1, \dots, m$.

Intuitions: Note that $W_n Y_n = G_n(\lambda_0) X_n \beta_0 + G_n(\lambda_0) \epsilon_n$.

- The chosen Q_n is correlated with $G_n X_n \beta_0$ but uncorrelated with ϵ_n , instrumenting the mean of $W_n Y_n$;
- $P_{jn} \epsilon_n$ is uncorrelated with ϵ_n , and thus it instruments the error of $W_n Y_n$, if P_{jn} is chosen such that $P_{jn} \epsilon_n$ is correlated with $G_n \epsilon_n$,
- suggesting that P_{jn} can be W_n , $W_n^2 - \frac{1}{n} \text{tr}(W_n^2) I_n$, etc.

Letting $Q_n = (X_n, Q_n)$, the **GMM estimator** of δ_0 is

$$\tilde{\delta}_n = \text{argmin } \mathbf{g}'_n(\delta) \Omega_n \mathbf{g}_n(\delta),$$

where $\mathbf{g}_n(\delta) = \{\epsilon'_n(\delta) Q_n, \epsilon'_n(\delta) P_{1n} \epsilon_n(\delta), \dots, \epsilon'_n(\delta) P_{mn} \epsilon_n(\delta)\}'$, and Ω_n is the weight matrix. The asymptotic VC matrix of $\tilde{\delta}_n$ is given as:

$$\text{AVar}(\tilde{\delta}_n) = (\Sigma'_n \Omega_n \Sigma_n)^{-1} (\Sigma'_n \Omega_n \Gamma_n \Omega_n \Sigma_n) (\Sigma'_n \Omega_n \Sigma_n)^{-1},$$

where $\Gamma_n = \text{Var}[\mathbf{g}_n(\delta_0)]$ and $\Sigma_n = -E[\frac{\partial}{\partial \delta'} \mathbf{g}_n(\delta_0)]$, with exact expressions:

$$\Sigma_n = \begin{pmatrix} \mathbb{Q}_n' X_n & \mathbb{Q}_n' G_n X_n \beta_0 \\ 0 & \sigma_0^2 \text{tr}(P_{1n}^s G_n) \\ \vdots & \vdots \\ 0 & \sigma_0^2 \text{tr}(P_{mn}^s G_n) \end{pmatrix}, \quad (28)$$

$$\Gamma_n = \begin{pmatrix} \sigma_0^2 \mathbb{Q}_n' \mathbb{Q}_n & \sigma_0^3 \gamma_0 \mathbb{Q}_n' \omega_{nm} \\ \sigma_0^3 \gamma_0 \omega_{nm}' \mathbb{Q}_n & \sigma_0^4 (\Lambda_{mn} + \kappa_0 \omega_{nm}' \omega_{nm}) \end{pmatrix}, \quad (29)$$

where $P_{kn}^s = P_{kn}' + P_{kn}$, $k = 1, \dots, m$, $\omega_{nm} = \{\text{diagv}(P_{1n}), \dots, \text{diagv}(P_{mn})\}$, and $\Lambda_{mn} = \{\text{tr}(P_{jn} P_{kn}^s), j, k = 1, \dots, m\}$.

- Simplest GMM: $\Omega_n = I_n$;
- Optimal GMM: $\Omega_n = \Gamma_n^{-1}$, not feasible as Γ_n contains σ_0^2 , γ_0 and κ_0 ;
- Feasible optimal GMM: $\Omega_n = \tilde{\Gamma}_n^{-1}$, where $\tilde{\Gamma}_n$ is a 'consistent estimate' of Γ_n , based on initial consistent estimate of δ_0 ;
- Best OPGMM: choose 'best' moment functions, . . . (see Lee, 2007a).

A simple choice for $\tilde{\Gamma}_n$ is $\tilde{\Gamma}_{2SLS}$ based on the 2SLS estimation:

- (i) let $Z_n = (X_n, W_n Y_n)$, and compute the 2SLSE of δ_0 :

$$\tilde{\delta}_{2SLS} = [Z_n' Q_n (Q_n' Q_n)^{-1} Q_n' Z_n]^{-1} Z_n' Q_n (Q_n' Q_n)^{-1} Q_n' Y_n,$$

- (ii) compute the 2SLE residuals $\tilde{\epsilon}_{2SLS} = A_n(\tilde{\lambda}_{2SLS}) - X_n \tilde{\beta}_{2SLS}$, and the sample variance, skewness and excess kurtosis of $\tilde{\epsilon}_{2SLS}$ to give consistent estimates of σ^2 , γ_0 and κ_0 .

The feasible OGMM estimator of δ_0 is

$$\tilde{\delta}_n^\circ = \operatorname{argmin} \mathbf{g}_n'(\delta) \tilde{\Gamma}_{2SLS}^{-1} \mathbf{g}_n(\delta),$$

and a consistent estimate of the asymptotic VC matrix of $\tilde{\delta}_n^\circ$ is

$$(\tilde{\Sigma}_{2SLS}' \tilde{\Gamma}_{2SLS}^{-1} \tilde{\Sigma}_{2SLS})^{-1}.$$

- For more details on GMM estimation of SLD model, see Lee (2007a).
- A thorough Monte Carlo comparison of QMLE and GMME are desirable, based on 'strong' or 'weak' instruments.

2.3.3. Testing statistical hypothesis

Tests for covariate effects. Let $\hat{V}_n = \hat{\mathcal{J}}_n^{-1} \hat{\mathcal{I}}_n \hat{\mathcal{J}}_n^{-1}$, where $\hat{\mathcal{J}}_n$ and $\hat{\mathcal{I}}_n$ are the plug-in estimates of $\mathcal{J}_n$ and $\mathcal{I}_n$. Letting $\hat{V}_{n\beta\beta}$ be the β - β block of $\hat{V}_n$, we obtain a Wald statistic for $c_0' \beta_0$, which is asymptotic $N(0, 1)$:

$$t_{\text{SLD}}(\beta_0) = \frac{c_0' \hat{\beta}_n - c_0' \beta_0}{\sqrt{c_0' \hat{V}_{n\beta\beta} c_0}} \xrightarrow{D} N(0, 1). \quad (30)$$

- The finite sample performance of $t_{\text{SLD}}(\beta_0)$ can be poor. This is because the QMLE $\hat{\lambda}_n$ is downward biased (Yang, 2015), which quickly passes to the QMLE of β_0 as seen from below:

$$\hat{\beta}_n = \tilde{\beta}_n(\hat{\lambda}_n) = \beta_0 + (\lambda - \hat{\lambda}_n)(X_n' X_n)^{-1} X_n' G_n X_n \beta_0 + o_p(1).$$

- As a result, the variance estimate $\hat{V}_{n\beta\beta}$ can also be unreliable.
- These are confirmed by the Monte Carlo results given in Liu and Yang (2015a), where improved inferences methods are provided (see Chapter 3 of the lecture notes).

Wald Test for spatial effect. Similarly, letting $\hat{V}_{n\lambda\lambda}$ be the λ - λ element of $\hat{V}_n$, we obtain a Wald statistic for λ , which is asymptotic $N(0, 1)$,

$$t_{\text{SLD}}(\lambda_0) = \frac{\hat{\lambda}_n - \lambda_0}{\sqrt{\hat{V}_{n\lambda\lambda}}}, \quad (31)$$

- The statistic $t_{\text{SLD}}(\lambda_0)$ can be used to test $H_0: \lambda = 0$.
- Finite sample property of $t_{\text{SLD}}(\lambda_0)$ is studied by Yang (2015),
- along with the improved tests based on bias-corrections.

LM tests for spatial effect. For testing $H_0: \lambda = 0$, LM tests are preferred as they require only the OLS estimates.

Anselin (1988) presents an LM test based on the expected information:

$$\text{LME}_{\text{SLD}} = \frac{\tilde{\epsilon}_n' W_n Y_n}{\tilde{\sigma}_n^2 \sqrt{\tilde{D}_n + T_n}}, \quad (32)$$

where $T_n = \text{tr}[(W_n + W_n')W_n]$, $\tilde{D}_n = \tilde{\sigma}_n^{-2}(W_n X_n \tilde{\beta}_n)' M_n W_n X_n \tilde{\beta}_n$, $\tilde{\beta}_n$ and $\tilde{\sigma}_n^2$ are the OLS estimates, and $\tilde{\epsilon}_n$ are the OLS residuals.

Paralleled with the OPG-based LM test (18), Born and Breitung (2011) also derived an OPG-based LM test for the lack of spatial lag dependence in the SLD model:

$$\text{LMO}_{\text{SLD}} = \frac{\tilde{\epsilon}_n' W_n Y_n}{\sqrt{(\tilde{\epsilon}_n \odot \tilde{\epsilon}_n)' (\tilde{\xi}_n \odot \tilde{\xi}_n)}}, \quad (33)$$

where $\tilde{\xi}_n = (W_n^l + W_n^{u'})\tilde{\epsilon}_n$, W_n^l and W_n^u are the lower and upper triangular matrices such that $W_n^l + W_n^u = W_n$, and $\text{LMO}_{\text{SLD}}|_{H_0} \xrightarrow{D} N(0, 1)$.

- Again, this test is robust against unknown heteroskedasticity, but may perform poorly in finite sample.
- Baltagi and Yang (2013a) introduced a standardized version of this test which has a much better finite sample performance.

Finite sample performance

Refer to Yang (2015) and Liu and Yang (2015b) for Monte Carlo results on the finite sample performance of the QMLEs and the finite sample performance of the t -statistics.

Empirical illustrations

Boston House Price. Refer to the earlier empirical illustration for the SED model for the details on the data.

Table 2.3. Estimation of SLD Model: Boston House Price

	QMLE	rse	t-Ratio	QMLE-bc	rse-bc	t-Ratio-bc
constant	14.5396	5.0687	2.8685	13.4682	5.0337	2.6756
crime	-0.0831	0.0300	-2.7678	-0.0819	0.0300	-2.7326
zoning	0.0448	0.0124	3.6167	0.0447	0.0124	3.6151
industry	0.0353	0.0555	0.6368	0.0361	0.0554	0.6506
charlesr	1.1181	0.7850	1.4243	1.0414	0.7835	1.3291
noxsq	-11.4788	3.5190	-3.2620	-11.1715	3.5104	-3.1824
rooms2	3.7066	0.3800	9.7543	3.7016	0.3794	9.7564
houseage	0.0021	0.0119	0.1733	0.0021	0.0119	0.1792
distance	-1.2240	0.1842	-6.6431	-1.2117	0.1838	-6.5921
access	0.2553	0.0604	4.2258	0.2529	0.0603	4.1921
taxrate	-0.0110	0.0034	-3.2225	-0.0109	0.0034	-3.2075
ptratio	-0.5349	0.1247	-4.2880	-0.5145	0.1242	-4.1427
blackpop	0.0078	0.0024	3.2216	0.0078	0.0024	3.1968
lowclass	-0.3679	0.0483	-7.6215	-0.3602	0.0481	-7.4952
σ^2	18.3358	2.2937	7.9938	18.2882	2.2845	8.0053
λ	0.3833	0.0399	9.6044	0.4020	0.0386	10.4204

2.4. SLE Model

The model. Adding both spatial lag and error (SLE) into the linear regression models, we have a more general spatial model:

$$Y_n = \lambda W_{1n} Y_n + X_n \beta + u_n, \quad u_n = \rho W_{2n} u_n + \epsilon_n, \quad (34)$$

which is commonly called the spatial autoregressive (SAR) model with autoregressive errors, or for short the SARAR(1,1) model.

2.4.1. QML estimation. Letting $A_n(\lambda) = I_n - \lambda W_{1n}$, $B_n(\rho) = I_n - \rho W_{2n}$, the model (34) has a reduced form: $B_n(\rho)A_n(\lambda)Y_n = B_n(\rho)X_n\beta + \epsilon_n$. The Jacobian of the inverse transformation ($\epsilon_n \rightarrow Y_n$) is thus $|B_n(\rho)A_n(\lambda)|$. The Gaussian log-likelihood function of $\theta = (\beta', \sigma^2, \lambda, \rho)'$ is

$$\begin{aligned} \ell_n(\theta) = & -\frac{n}{2} \log(2\pi\sigma^2) + \log |A_n(\lambda)| + \log |B_n(\rho)| \\ & - \frac{1}{2\sigma^2} [Y_n(\delta) - X_n(\rho)\beta]' [Y_n(\delta) - X_n(\rho)\beta], \end{aligned} \quad (35)$$

where $X_n(\rho) = B_n(\rho)X_n$, $Y_n(\delta) = B_n(\rho)A_n(\lambda)Y_n$, and $\delta = (\lambda, \rho)'$.

Maximizing $\ell_n(\theta)$ gives the MLE $\hat{\theta}_n$ of θ if $\epsilon_n \sim N(0, \sigma^2 I_n)$, otherwise the QMLE of θ . The score function has the form:

$$S_n(\theta) = \frac{\partial}{\partial \theta} \ell_n(\theta) = \begin{cases} \frac{1}{\sigma^2} X_n'(\rho) \epsilon_n(\beta, \delta), \\ \frac{1}{2\sigma^4} \epsilon_n'(\beta, \lambda) \epsilon_n(\beta, \delta) - \frac{n}{2\sigma^2}, \\ \frac{1}{\sigma^2} Y_n' W_{1n}' B_n'(\rho) \epsilon_n(\beta, \delta) - \text{tr}[G_{1n}(\lambda)], \\ \frac{1}{\sigma^2} \epsilon_n'(\beta, \delta) G_{2n}(\rho) \epsilon_n(\beta, \delta) - \text{tr}[G_{2n}(\rho)], \end{cases} \quad (36)$$

where $\epsilon_n(\beta, \delta) = Y(\delta) - X_n(\rho)\beta$, $G_{1n}(\lambda) = W_{1n}A_n^{-1}(\lambda)$, and $G_{2n}(\rho) = W_{2n}B_n^{-1}(\rho)$.

Given δ , solving the first two components of the score-based estimation equations $S_n(\theta) = 0$, we obtain the constrained (Q)MLEs of β and σ^2 :

$$\tilde{\beta}_n(\delta) = [X_n'(\rho)X_n(\rho)]^{-1} X_n'(\rho) Y_n(\delta), \quad (37)$$

$$\tilde{\sigma}_n^2(\delta) = \frac{1}{n} Y_n'(\delta) M_n(\rho) Y_n(\delta), \quad (38)$$

where $M_n(\rho) = I_n - B_n(\rho)X_n[X_n'B_n'(\rho)B_n(\rho)X_n]^{-1}X_n'B_n'(\rho)$. Substituting $\tilde{\beta}_n(\delta)$ and $\tilde{\sigma}_n^2(\delta)$ back into (35), we obtain the concentrated loglikelihood for δ :

$$\ell_n^c(\delta) = -\frac{n}{2}[\ln(2\pi) + 1] - \frac{n}{2}\ln(\hat{\sigma}_n^2(\delta)) + \ln|A_n(\lambda)| + \ln|B_n(\rho)|. \quad (39)$$

Maximizing (39) gives the QMLE $\hat{\delta}_n$ of δ , and thus the QMLEs of β and σ^2 as $\hat{\beta}_n \equiv \tilde{\beta}_n(\hat{\delta}_n)$ and $\hat{\sigma}_n^2 \equiv \tilde{\sigma}_n^2(\hat{\delta}_n)$. Write $\hat{\theta}_n = (\hat{\beta}_n', \hat{\sigma}_n^2, \hat{\delta}_n')'$. The concentrated score function upon dividing by n is,

$$\tilde{\psi}_n(\delta) = \begin{cases} -\frac{1}{n}\text{tr}(\mathbf{G}_{1n}(\lambda)) + \frac{Y_n'(\delta)\mathbf{M}_n(\rho)\bar{\mathbf{B}}_n(\delta)Y_n(\delta)}{Y_n'(\delta)\mathbf{M}_n(\rho)Y_n(\delta)}, \\ -\frac{1}{n}\text{tr}(\mathbf{G}_{2n}(\rho)) + \frac{Y_n'(\delta)\mathbf{M}_n(\rho)\mathbf{G}_{2n}(\rho)\mathbf{M}_n(\rho)Y_n(\delta)}{Y_n'(\lambda)\mathbf{M}_n(\rho)Y_n(\delta)}, \end{cases} \quad (40)$$

where $\bar{\mathbf{B}}_n(\delta) = \mathbf{B}_n(\rho)\mathbf{G}_{1n}(\lambda)\mathbf{B}_n^{-1}(\rho)$. These are the key expressions for deriving the score-based tests for the spatial effects, and for performing bias-correction to be introduced in the subsequent chapters.

Asymptotic properties of the QMLEs. Jin and Lee (2013) shows that under some regularity conditions, $\hat{\theta}_n \xrightarrow{p} \theta_0$, and

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{D} N(\theta_0, \lim_{n \rightarrow \infty} n\mathcal{J}_n^{-1}\mathcal{I}_n\mathcal{J}_n^{-1}), \quad (41)$$

where $\mathcal{J}_n = -E[\frac{\partial}{\partial \theta'} S_n(\theta_0)]$ and $\mathcal{I}_n = \text{Var}[S_n(\theta_0)]$, having the expressions:

$$\mathcal{J}_n = \begin{pmatrix} \frac{1}{\sigma_0^2} X_n' B_n' B_n X_n & 0 & \frac{1}{\sigma_0} X_n' B_n' \mu_n & 0 \\ \sim & \frac{n}{2\sigma_0^4} & \frac{1}{\sigma_0^2} \text{tr}(\bar{B}_n) & \frac{1}{\sigma_0^2} \text{tr}(G_{2n}) \\ \sim & \sim & \mu_n' \mu_n + \text{tr}(\bar{B}_n^s \bar{B}_n) & \text{tr}(G_{2n}^s \bar{B}_n) \\ \sim & \sim & \sim & \text{tr}(G_{2n}^s G_{2n}) \end{pmatrix} \quad (42)$$

$$\mathcal{I}_n = \mathcal{J}_n + \begin{pmatrix} 0 & \frac{\gamma_0}{2\sigma_0^3} X_n' B_n' \iota_n & \frac{\gamma_0}{\sigma_0} X_n' B_n' \bar{b}_n & \frac{\gamma_0}{\sigma_0} X_n' B_n' g_{2n} \\ \sim & \frac{n\kappa_0}{4\sigma_0^4} & \frac{\kappa_0}{2\sigma_0^2} \text{tr}(\bar{B}_n) + \frac{\gamma_0}{2\sigma_0^2} \iota_n' \mu_n & \frac{\kappa_0}{2\sigma_0^2} \text{tr}(G_{2n}) \\ \sim & \sim & \kappa_0 \bar{b}_n' \bar{b}_n + 2\gamma_0 \bar{b}_n' \mu_n & \kappa_0 g_{2n}' \bar{b}_n + \gamma_0 g_{2n}' \mu_n \\ \sim & \sim & \sim & \kappa_0 g_{2n}' g_{2n} \end{pmatrix} \quad (43)$$

where ι_n , γ_0 and κ_0 are defined in the earlier sections, $\mu_n = \sigma_0^{-1} B_n G_{1n} X_n \beta_0$, $\bar{b}_n = \text{diag}(\bar{B}_n)$, $g_{2n} = \text{diagv}(G_{2n})$, $\bar{B}_n^s = \bar{B}_n + \bar{B}_n'$ and $G_{2n}^s = G_{2n} + G_{2n}'$.

- The results (41)-(43) form the base for statistical inferences.
- $\mathcal{J}_n$ is estimated by the plug-in estimator $\mathcal{J}_n(\hat{\theta}_n)$, or by $-\frac{\partial}{\partial \theta} S_n(\theta)|_{\theta=\hat{\theta}_n}$.
- $\mathcal{I}_n$ can be consistently estimated by the 'plug-in' method, i.e., plugging in $\hat{\theta}_n$, $\hat{\gamma}_n$ and $\hat{\kappa}_n$, where $\hat{\gamma}_n$ and $\hat{\kappa}_n$ are the sample skewness and excess kurtosis of the estimated errors $\hat{\epsilon}_n = Y_n(\hat{\delta}_n) - X_n(\hat{\rho}_n)\hat{\beta}_n$.

2.4.2. GMM estimation

Lee and Liu (2010) extend the GMM framework for SLD model to high-order spatial linear regression model, including the SLE model discussed above. The set of moment functions for the SLE or SARAR(1,1) model remains in a similar form as that for the SLD model:

$$\mathbf{g}_n(\delta) = \{\epsilon'_n(\delta)\mathbf{Q}_n, \epsilon'_n(\delta)\mathbf{P}_{1n}\epsilon_n(\delta), \dots, \epsilon'_n(\delta)\mathbf{P}_{mn}\epsilon_n(\delta)\}'$$

except that now $\delta = (\beta', \lambda, \rho)'$ and $\epsilon_n(\delta) = \mathbf{B}_n(\rho)[\mathbf{A}_n(\lambda)\mathbf{Y}_n - \mathbf{X}_n\beta]$.

Now, letting again $\Sigma_n = -E[\frac{\partial}{\partial \delta'} \mathbf{g}_n(\delta_0)]$ and $\Gamma_n = \text{Var}[\mathbf{g}_n(\delta_0)]$, we have

$$\Sigma_n = \begin{pmatrix} \mathbf{Q}'_n \mathbf{X}_n & \mathbf{Q}'_n \mathbf{B}_n \mathbf{G}_{1n} \mathbf{X}_n \beta_0 & 0 \\ 0 & \sigma_0^2 \text{tr}(\mathbf{P}_{1n}^s \mathbf{B}_n \mathbf{G}_{1n} \mathbf{B}_n^{-1}) & \sigma_0^2 \text{tr}(\mathbf{P}_{1n}^s \mathbf{G}_{2n}) \\ \vdots & \vdots & \vdots \\ 0 & \sigma_0^2 \text{tr}(\mathbf{P}_{mn}^s \mathbf{B}_n \mathbf{G}_{1n} \mathbf{B}_n^{-1}) & \sigma_0^2 \text{tr}(\mathbf{P}_{mn}^s \mathbf{G}_{2n}) \end{pmatrix}, \quad (44)$$

and Γ_n identical in expression as that given in (29).

The feasible OGMM estimate of δ_0 is thus

$$\tilde{\delta}_n = \operatorname{argmin} \mathbf{g}'_n(\delta) \tilde{\Gamma}_n^{-1} \mathbf{g}_n(\delta)$$

where $\tilde{\Gamma}_n$ is a consistent estimate of Γ_n , which can be obtained based on the generalized 2SLS estimate of Kelejian and Prucha (1998):

- (i) let $Z_n = (X_n, W_n Y_n)$, and compute the G2SLSE of δ_0 :

$$\begin{aligned} \tilde{\delta}_{\text{G2SLS}} &= [Z'_n B'_n(\tilde{\rho}_n) Q_n (Q'_n Q_n)^{-1} Q'_n B_n(\tilde{\rho}_n) Z_n]^{-1} \\ &\quad Z'_n B'_n(\tilde{\rho}_n) Q_n (Q'_n Q_n)^{-1} Q'_n B_n(\tilde{\rho}_n) Y_n, \end{aligned}$$

where $\tilde{\rho}_n$ is a consistent initial estimator of ρ .

- (ii) compute the G2SLE residuals $\tilde{\epsilon}_{\text{G2SLS}} = A_n(\tilde{\lambda}_{\text{G2SLS}}) - X_n \tilde{\beta}_{\text{G2SLS}}$, and the sample variance, skewness and excess kurtosis of $\tilde{\epsilon}_{\text{G2SLS}}$ to give consistent estimates of σ_0^2 , γ_0 and κ_0 .

Finally, the IV matrix Q_n can be taken as fixed subset of linearly independent columns of $\{X_n, W_{1n} X_n, W_{1n}^2 X_n, W_{2n} X_n, W_{2n}^2 X_n\}$, and the P_{jn} matrices can be constructed similarly to these for the SLD model.

2.4.3. Testing statistical hypotheses

Let $V_{n\beta}$ and $V_{n\delta}$ be the submatrices of $\mathcal{J}_n^{-1}\mathcal{I}_n\mathcal{J}_n^{-1}$ corresponding to β and $\delta = (\lambda, \rho)'$, respectively. Let $\hat{V}_{n\beta}$ and $\hat{V}_{n\delta}$ be their estimates

- An asymptotic t -statistic for inferences for $c_0'\beta_0$ is thus,

$$t_{\text{SLE}}(\beta_0) = \frac{c_0'\hat{\beta}_n - c_0'\beta_0}{\sqrt{c_0'\hat{V}_{n\beta}c_0}} \sim N(0, 1), \quad (45)$$

where c_0 represents a linear contrast. A multivariate version can be constructed (see the corresponding lecture note).

- Similarly, a chi-squared statistic for inferences for δ is

$$T_{\text{SLE}}(\delta_0) = (\hat{\delta}_n - \delta_0)'\hat{V}_{n\lambda}^{-1}(\hat{\delta}_n - \delta_0) \sim \chi_2^2. \quad (46)$$

This test statistic can be used to test $H_0: \delta = 0$. One directional tests for λ (allowing ρ) and for ρ (allowing λ) can easily be formulated.

- For testing $H_0: \delta = 0$, LM tests are preferred as they require only the estimates of the null model, which in this case is the OLS regression.

El-based LM test.

Anselin (1988) gives a El-based LM test for testing $H_0: \delta = 0$:

$$\text{LME}_{\text{SLE}} = \frac{1}{\tilde{\sigma}_n^4} \begin{pmatrix} \tilde{\epsilon}'_n W_{1n} Y_n \\ \tilde{\epsilon}'_n W_{2n} \tilde{\epsilon}_n \end{pmatrix}' \begin{pmatrix} T_{1n} + \tilde{D}_n & T_{3n} \\ T_{3n} & T_{2n} \end{pmatrix}^{-1} \begin{pmatrix} \tilde{\epsilon}'_n W_{1n} Y_n \\ \tilde{\epsilon}'_n W_{2n} \tilde{\epsilon}_n \end{pmatrix}, \quad (47)$$

- $T_{rn} = \text{tr}[(W_{rn} + W'_{rn})W_{rn}]$, $r = 1, 2$,
- $T_{3n} = \text{tr}[(W_{2n} + W'_{2n})W_{1n}]$,
- $\tilde{D}_n$ is defined in (32).

OPG-based LM test.

Born and Breitung (2011) give an OPG-based test for testing $H_0: \delta = 0$:

$$\text{LMO}_{\text{SLE}} = \begin{pmatrix} \tilde{\epsilon}'_n W_{1n} Y_n \\ \tilde{\epsilon}'_n W_{2n} \tilde{\epsilon}_n \end{pmatrix}' \begin{pmatrix} \tilde{\epsilon}_n^2{}' \tilde{\xi}_{1n}^2 & \tilde{\epsilon}_n^2{}' (\tilde{\xi}_{1n} \odot \tilde{\xi}_{2n}) \\ \tilde{\epsilon}_n^2{}' (\tilde{\xi}_{1n} \odot \tilde{\xi}_{2n}) & \tilde{\epsilon}_n^2{}' \tilde{\xi}_{2n}^2 \end{pmatrix}^{-1} \begin{pmatrix} \tilde{\epsilon}'_n W_{1n} Y_n \\ \tilde{\epsilon}'_n W_{2n} \tilde{\epsilon}_n \end{pmatrix}, \quad (48)$$

- *square* of a vector, e.g., $\tilde{\epsilon}_n^2 = \tilde{\epsilon}_n \odot \tilde{\epsilon}_n$,
- W_{rn}^u and W_{rn}^l : the upper and lower triangular matrices of W_{rn} , $r = 1, 2$,
- $\tilde{\xi}_{1n} = (W_{1n}^{u'} + W_{1n}^l)\tilde{\epsilon}_n + M_n W_n X_n \tilde{\beta}_n$,
- $\tilde{\xi}_{2n} = (W_{2n}^{u'} + W_{2n}^l)\tilde{\epsilon}_n$.

- Both LM tests do not take into account the estimation of β and σ^2 , and hence may suffer from the problems of size distortion due mainly to the lack of centering and rescaling.
- Baltagi and Yang (2013b) introduce a standardized version of it which performs much better in finite sample (see Chapter 3 of the notes).

Finite sample performance

Table 2.4 Empirical Sizes: Two-Sided Tests of $H_0 : \beta_1 = \beta_2$ in SLE Model
Group Interaction, REG2, $\sigma = 1, \lambda = 0.5$; Test: $1 = t_{SLE}, 2 = t_{SLE}^{bc}, 3 = t_{SLE}^{bc2}$

ρ Test	10%	5%	1%	10%	5%	1%	10%	5%	1%	10%	5%	1%	10%	5%	1%	10%	5%	1%	
	Normal			Nor-Mixture			Lognormal			Normal			Nor-Mixture			Lognormal			
	$n = 50$									$n = 200$									
.50	1	.197	.115	.040	.201	.122	.044	.197	.122	.040	.141	.078	.022	.140	.082	.028	.131	.078	.021
	2	.120	.068	.020	.123	.073	.023	.146	.084	.028	.113	.056	.014	.117	.061	.017	.116	.061	.015
	3	.115	.062	.017	.119	.068	.023	.128	.074	.024	.105	.050	.011	.107	.056	.016	.106	.052	.013
.00	1	.191	.110	.031	.177	.099	.028	.191	.114	.037	.150	.089	.026	.137	.075	.016	.138	.084	.020
	2	.111	.054	.015	.100	.054	.016	.117	.065	.021	.104	.055	.012	.116	.061	.014	.124	.066	.017
	3	.098	.047	.012	.095	.046	.013	.100	.055	.018	.097	.050	.010	.102	.051	.010	.105	.052	.012
-.50	1	.182	.104	.030	.162	.085	.023	.177	.100	.034	.127	.072	.020	.120	.061	.013	.119	.063	.013
	2	.097	.049	.013	.085	.043	.010	.102	.059	.019	.115	.066	.017	.122	.063	.015	.135	.074	.019
	3	.100	.048	.011	.091	.052	.009	.092	.046	.014	.105	.060	.013	.095	.046	.009	.091	.052	.009

Empirical illustration. (Boston House Price.) Refer to the empirical applications for the SED model for the detailed description of the data.

Table 2.5. Estimation of SLE Model: Boston House Price

	QMLE	rse	t-Ratio	QMLE-bc	rse-bc	t-Ratio-bc
constant	27.8251	5.9648	4.6649	27.8251	5.9648	4.6649
crime	-0.1301	0.0279	-4.6660	-0.1301	0.0279	-4.6660
zoning	0.0390	0.0141	2.7763	0.0390	0.0141	2.7763
industry	-0.0071	0.0723	-0.0976	-0.0071	0.0723	-0.0976
charlesr	-0.3970	0.8775	-0.4524	-0.3970	0.8775	-0.4524
noxsq	-18.5059	5.1765	-3.5750	-18.5059	5.1765	-3.5750
rooms2	4.2958	0.3680	11.6746	4.2958	0.3680	11.6746
houseage	-0.0240	0.0139	-1.7300	-0.0240	0.0139	-1.7300
distance	-1.6254	0.2897	-5.6097	-1.6254	0.2897	-5.6097
access	0.3177	0.0756	4.2047	0.3177	0.0756	4.2047
taxrate	-0.0131	0.0036	-3.6294	-0.0131	0.0036	-3.6294
ptratio	-0.6201	0.1513	-4.0984	-0.6201	0.1513	-4.0984
blackpop	0.0104	0.0031	3.3818	0.0104	0.0031	3.3818
lowclass	-0.4234	0.0521	-8.1350	-0.4234	0.0521	-8.1350
σ^2	15.0920	1.8042	8.3648	15.0920	1.8042	8.3648
λ	0.0569	0.0679	0.8381	0.0506	0.0679	0.7456
ρ	0.6595	0.0593	11.1309	0.6893	0.0593	11.6339

Appendix: Basics on Consistency and Asymptotic Normality

This appendix presents some basics for establishing consistency and asymptotic normality of a QML estimator or an M-estimator.

Consistency. Let δ be the parameter vector in the concentrated loglikelihood $\ell_n^c(\delta)$, obtained from the full loglikelihood $\ell_n(\theta)$ by replacing β and σ^2 by their constrained QMLEs.

- Let δ_0 be the true value of θ which takes values in the parameter space Θ . The expectation operator 'E' corresponds the true θ_0 , or the true distribution.
- It is only necessary to prove the consistency of $\hat{\delta}_n$ as the consistency of $\hat{\beta}_n$ and $\hat{\sigma}_n$ immediately follow if $\text{plim} \frac{1}{n} X_n' X_n$ exists and is positive definite (p.d.).
- Define $\bar{\ell}_n(\theta) = E[\ell_n(\theta)]$, the expected loglikelihood at a general θ .
- Define

$$\bar{\ell}_n^c(\delta) = \max_{\beta, \sigma^2} \bar{\ell}_n(\theta).$$

The consistency of $\hat{\delta}_n$ follows, i.e., $\hat{\delta}_n \xrightarrow{P} \delta_0$, if

(a) $\sup_{\delta: d(\delta, \delta_0) \geq \varepsilon} \bar{\ell}_n^c(\delta) < \bar{\ell}_n^c(\delta_0)$, for every $\varepsilon > 0$,

(b) $\sup_{\theta \in \Theta} \frac{1}{n} |\ell_n^c(\delta) - \bar{\ell}_n^c(\delta)| \xrightarrow{P} 0$.

The condition (a) is the so-called identification uniqueness condition for δ_0 which ensures that $\bar{\ell}_n^c(\delta)$ has a identifiably unique maximizer that converges to δ_0 .

This together with condition (b) ensures that the difference between the maximizer $\hat{\theta}_n$ of $\ell_n^c(\delta)$ and the maximizer of $\bar{\ell}_n^c(\delta)$ gets smaller and smaller as n goes large. See White (1994, Theorem 3.4) or van der Vaar (1998, Theorem 5.7).

For detailed applications of these theorems, see Liu and Yang (2015a) for the case of SED model, Lee (2004) for SLD model and Jin and Lee (2013) for the SLE model.

Asymptotic normality. Assume the QMLE $\hat{\theta}_n$ which maximizes $\ell_n(\theta)$, can be equivalently obtained by solving the estimating equation $S_n(\theta) = 0$, where $S_n(\theta) = \frac{\partial}{\partial \theta} \ell_n(\theta)$ is the score function. Assume $\hat{\theta}_n \xrightarrow{P} \theta_0$. Then, by the **mean value theorem**, we have

$$0 = \frac{1}{\sqrt{n}} S_n(\hat{\theta}_n) = \frac{1}{\sqrt{n}} S_n(\theta_0) + \left[\frac{1}{n} \frac{\partial}{\partial \theta'} S_n(\theta^*) \right] \sqrt{n}(\hat{\theta}_n - \theta_0),$$

where θ^* lies elementwise between $\hat{\theta}_n$ and θ_0 . The asymptotic normality of $\hat{\theta}_n$ follows if

- (a) $\frac{1}{\sqrt{n}} S_n(\theta_0) \xrightarrow{D} N(0, \lim_{n \rightarrow \infty} \frac{1}{n} \mathcal{I}_n)$,
- (b) $\frac{1}{n} \left[\frac{\partial}{\partial \theta'} S_n(\theta^*) - \frac{\partial}{\partial \theta'} S_n(\theta_0) \right] \xrightarrow{P} 0$, and
- (c) $\frac{1}{n} \left[\frac{\partial}{\partial \theta'} S_n(\theta_0) - E\left(\frac{\partial}{\partial \theta'} S_n(\theta_0)\right) \right] \xrightarrow{P} 0$.

As the elements of $S_n(\theta_0)$ are typically linear or quadratic forms in ϵ_n due to the use of Gaussian quasi likelihood, the result (a) can be proved using the central limit theorem (CLT) for linear-quadratic forms of Kelejian and Prucha (2001), and the Cramér-Wold device.

The conditions (b) and (c) can be proved by applying some law of large numbers (LLN). See Liu and Yang (2015a) for the case of SED model, Lee (2004) for the case of SLD model, and Jin and Lee (2013) for the case of SLE model.

Clearly, if the above conditions hold, then

$$\begin{aligned} 0 &= \frac{1}{\sqrt{n}} S_n(\theta_0) + \left[\frac{1}{n} E\left(\frac{\partial}{\partial \theta'} S_n(\theta_0)\right) \right] \sqrt{n}(\hat{\theta}_n - \theta_0) + o_p(1) \\ &= \frac{1}{\sqrt{n}} S_n(\theta_0) - \left[\frac{1}{n} \mathcal{J}_n(\theta_0) \right] \sqrt{n}(\hat{\theta}_n - \theta_0) + o_p(1) \end{aligned}$$

It follows that $\sqrt{n}(\hat{\theta}_n - \theta_0) = \left[\frac{1}{n} \mathcal{J}_n(\theta_0) \right]^{-1} \frac{1}{\sqrt{n}} S_n(\theta_0) + o_p(1)$. It follows that

$$\text{Var}[\sqrt{n}(\hat{\theta}_n - \theta_0)] = \left[\frac{1}{n} \mathcal{J}_n(\theta_0) \right]^{-1} \text{Var}\left[\frac{1}{\sqrt{n}} S_n(\theta_0)\right] \left[\frac{1}{n} \mathcal{J}_n(\theta_0) \right]^{-1}.$$

If further $E[S_n(\theta_0)] = 0$ or at least $\frac{1}{n} E[S_n(\theta_0)] \rightarrow 0$, then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{D} N(0, \lim_{n \rightarrow \infty} n \mathcal{J}_n^{-1} \mathcal{I}_n \mathcal{J}_n^{-1}).$$

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Table 1. Monte Carlo Mean[rmse](sd) of Estimators of λ in SLD Model

λ	n	$\hat{\lambda}_n$	$\hat{\lambda}_n^{bc2}$	$\hat{\lambda}_n^{bc3}$
Queen Contiguity, Normal Errors				
.50	50	.411 [.195](.174)	.492 .175	.497 .175
	100	.459 [.123](.116)	.498 .117	.500 .117
	200	.480 [.078](.076)	.499 .075	.499 .075
	500	.493 [.049](.048)	.501 .048	.501 .048
.25	50	.163 [.222](.204)	.242 .209	.246 .210
	100	.212 [.146](.140)	.248 .142	.250 .143
	200	.231 [.094](.092)	.250 .093	.250 .093
	500	.242 .060	.250 .060	.250 .060
.00	50	-.078 [.229](.216)	-.006 .224	-.003 .226
	100	-.034 [.157](.153)	-.002 .156	-.001 .157
	200	-.018 [.106](.104)	-.000 .105	.000 .105
	500	-.008 [.068](.067)	-.000 .068	-.000 .068
-.25	50	-.317 [.233](.223)	-.255 .236	-.254 .237
	100	-.279 [.164](.161)	-.253 .166	-.253 .166
	200	-.266 [.112](.111)	-.252 .112	-.251 .112
	500	-.256 [.073](.072)	-.250 .073	-.250 .073

Table 2. Monte Carlo Mean[rmse](sd) of Estimators of λ in SLD Model

λ	n	$\hat{\lambda}_n$	$\hat{\lambda}_n^{bc2}$	$\hat{\lambda}_n^{bc3}$
Queen Contiguity, Normal Mixture Errors				
.50	50	.420 [.182](.164)	.494 .165	.498 .165
	100	.462 [.120](.114)	.499 .114	.500 .114
	200	.482 [.076](.074)	.500 .074	.500 .074
	500	.492 [.049](.048)	.500 .048	.500 .048
.25	50	.169 [.207](.190)	.241 .195	.244 .195
	100	.213 [.140](.135)	.248 .136	.249 .137
	200	.230 [.092](.090)	.249 .090	.249 .090
	500	.242 .060	.250 .060	.250 .060
.00	50	-.070 [.217](.206)	-.004 .213	-.002 .214
	100	-.032 [.150](.147)	-.002 .150	-.001 .150
	200	-.018 [.104](.103)	-.001 .103	-.001 .103
	500	-.008 [.068](.067)	-.001 .067	-.001 .067
-.25	50	-.314 [.223](.213)	-.258 .224	-.257 .225
	100	-.275 [.155](.153)	-.251 .157	-.250 .157
	200	-.263 [.111](.110)	-.249 .112	-.249 .112
	500	-.257 .072	-.251 .072	-.251 .072

Table 3. Monte Carlo Mean[rmse](sd) of Estimators of λ in SLD Model

λ	n	$\hat{\lambda}_n$	$\hat{\lambda}_n^{bc2}$	$\hat{\lambda}_n^{bc3}$
Queen Contiguity, Lognormal Errors				
.50	50	.426 [.163](.146)	.491 .146	.493 .146
	100	.465 [.110](.105)	.498 .105	.498 .105
	200	.482 [.072](.069)	.499 .069	.499 .069
	500	.491 [.047](.046)	.499 .046	.499 .046
.25	50	.179 [.185](.171)	.241 .174	.244 .174
	100	.216 [.128](.124)	.247 [.126](.125)	.248 [.126](.125)
	200	.232 [.087](.085)	.249 .085	.249 .085
	500	.242 [.058](.057)	.249 .057	.249 .057
.00	50	-.067 [.198](.186)	-.011 [.192](.191)	-.008 .192
	100	-.029 [.139](.136)	-.003 .138	-.002 .138
	200	-.017 [.099](.097)	-.002 .098	-.001 .098
	500	-.007 [.065](.064)	-.000 .065	.000 .065
-.25	50	-.307 [.199](.191)	-.258 .198	-.256 .199
	100	-.272 [.142](.140)	-.252 .144	-.251 .144
	200	-.264 [.105](.104)	-.251 .105	-.250 .105
	500	-.256 .070	-.250 .070	-.250 .070